Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Phase Ib, Open-label, Multi-center Study to Characterize the Safety, Tolerability, and Preliminary Efficacy of TNO155 in Combination With Spartalizumab or Ribociclib in Selected Malignancies **Short Title/Acronym:** Phase Ib Study of TNO155 in Combination With Spartalizumab or Ribociclib **Posting ID:** 986228741632739140 **Protocol Number:** CTNO155B12101 **NCT Number:** NCT04000529 **Sponsor/Company Name:** Novartis Pharmaceuticals / Novartis **Investigational Product:** TNO155 (SHP2 inhibitor) in combination with Spartalizumab (anti-PD-1) or Ribociclib (CDK4/6 inhibitor; Trade Name: Kisqali, ATC Code: L01EF02) **Phase of Development:** Phase Ib (PHASE1) **Medical Monitor:** Novartis Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize the safety and tolerability of TNO155 in combination with spartalizumab and of TNO155 in combination with ribociclib, and to identify the Maximum Tolerated Dose (MTD) and/or recommended regimen (dose and schedule) for each combination. **Secondary Objectives:** To characterize the pharmacokinetic (PK) profile of TNO155, spartalizumab, and ribociclib when administered as combinations. To evaluate the preliminary anti-tumor activity (e.g., ORR, DCR, PFS) of the respective combinations.

2.2 Background & Rationale To address the need for improved targeted therapeutic options in advanced solid tumors driven by RTK and MAPK pathways. Data from preclinical models have demonstrated that combining TNO155 with spartalizumab or ribociclib shifts KRAS to the inactive GDP-bound state, demonstrating superior anti-tumor activity compared to single-agent administration. These combinations aim to overcome resistance and provide clinical benefit to patients with advanced malignancies who have progressed on standard-of-care (SOC) therapies.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Dose Escalation and Dose Expansion) **Control Type:** Active combinations / Parallel Assignment **Blinding:** Open-label **Randomization:** Non-Randomized

3.2 Planned Enrollment & Duration Target N: 122 participants **Number of Sites:** Multi-center (Global) **Estimated Study Start:** Jul-2019 **Trial Status / Primary Completion:** Terminated

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. (For Japan only: written consent is necessary both from the patient and his/her legal representative if he/she is under the age of 20 years). 2. Eastern Cooperative Oncology Group (ECOG) performance status ≤ 1. 3. Patients with advanced solid tumors (e.g., Non-small Cell Lung Carcinoma, Head and Neck Squamous Cell Carcinoma, Esophageal SCC, Gastrointestinal Stromal Tumors, Colorectal Cancer) with evaluable disease per RECIST v1.1 after progression on or intolerance to platinum-containing combination chemotherapy or standard-of-care.

4.2 Exclusion Criteria 1. Pregnant or breast-feeding (lactating) women. 2. History of severe hypersensitivity reactions to other monoclonal antibodies (mAbs). 3. Active, known, or suspected autoimmune disease, or a history of/current interstitial lung disease or pneumonitis grade ≥ 2. 4. Human Immunodeficiency Virus (HIV) infection, unless on antiviral therapy with an undetectable viral load.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: TNO155 administered orally (e.g., 60 mg/day or 40 mg/day on a 2-weeks-on/1-week-off schedule) in combination with either Spartalizumab (300 mg Q3W, intravenously) or Ribociclib (Trade Name: Kisqali; 200 mg/day, orally). **Route of Administration:** Oral (TNO155, Ribociclib) and Intravenous (Spartalizumab). **Duration of Treatment:** Administered in 21-day or 28-day cycles until unacceptable toxicity, progressive disease, or discontinuation at the discretion of the Investigator.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease-related symptoms and adverse events. **Prohibited:** Concurrent investigational anticancer therapies, other targeted therapies, or immunosuppressive medications that might interfere with spartalizumab efficacy or exacerbate toxicities.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Baseline Imaging, Labs
Baseline / C1D1	Day 1	Treatment Initiation, PK Blood Draws, Vital Signs, ECOG

Assessment | | **Interim Cycles** | Every 21 or 28 Days | Safety Labs, Efficacy Assessment (RECIST v1.1), PK/PD monitoring, AE tracking | | **End of Study** | Variable | Final Tumor Imaging, Exit Interview, IP Accountability, Survival Follow-up |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. **DLT incidence:** Incidence of dose limiting toxicities (DLTs) during the first cycle of combination treatment during the dose escalation part to identify the Maximum Tolerated Dose (MTD) and recommended dose. 2. **AE and SAE incidence:** Incidence and severity of adverse events (AEs) and serious adverse events (SAEs) as per CTCAE v5.0. 3. **Dose tolerability:** Dose interruptions, reductions and dose intensity.

7.2 Secondary & Exploratory Endpoints 1. Pharmacokinetic (PK) profiles: $Cmax$, $Tmax$, and $AUClast$ of TNO155, spartalizumab, and ribociclib. 2. Preliminary anti-tumor activity: Objective Response Rate (ORR), Disease Control Rate (DCR), Progression-Free Survival (PFS), and Duration of Response (DOR) evaluated using RECIST version 1.1.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of Treatment-Related Adverse Events (TRAEs), including AST/ALT elevations, anemia, thrombocytopenia, and peripheral edema. **Serious Adverse Events (SAEs):** Protocol-mandated rapid reporting for any fatal or severe events, dose reductions, or discontinuations (e.g., tracking pneumonitis, severe hypersensitivity). **Data Monitoring Committee (DMC):** Internal and independent dose-escalation review committees to adjudicate DLTs and approve cohort progression.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety set for all patients receiving at least one dose of the investigational combination; Efficacy set for patients with measurable disease at baseline. **Sample Size Justification:** Based on standard Phase Ib dose-escalation designs (e.g., 3+3 or Bayesian logistic regression models) to accurately estimate the MTD, followed by expansion cohorts. **Handling of Missing Data:** Standard oncology data handling practices for censored progression-free survival (PFS) and duration of response (DOR) utilizing Kaplan-Meier estimates.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine site visits and centralized medical review to assess adherence to the dose-escalation protocol and RECIST criteria. **Audit Trail:** Robust electronic Case Report Form (eCRF) tracking with source data verification.

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11. Study Identifiers & Governance

Posting ID: 986228741632739140 **Primary ID:** CTNO155B12101 **Registry Number:** NCT04000529 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Phase Ib Study of TNO155 in Combination With Spartalizumab or Ribociclib in Selected Malignancies **Official Regulatory Title:** A Phase Ib, Open-label, Multi-center Study to Characterize the Safety, Tolerability, and Preliminary Efficacy of TNO155 in Combination With Spartalizumab or Ribociclib in Selected Malignancies

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12. Clinical Framework (Oncology Specific)

Primary Condition: Non-small Cell Lung Carcinoma, Head and Neck Squamous Cell Carcinoma, Esophageal SCC, Gastrointestinal Stromal Tumors, Colorectal Cancer **Condition Ontology & Identifiers:** Carcinoma (Preferred UMLS Name: Malignancy; Semantic Type: Neoplastic Process; MeSH Code: D002277; SNOMED ID: 1187425009; HPO Code: HP:0030731) **Condition Definition:** A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas (NCI Thesaurus). **Diagnosis Threshold:** Progression on or intolerance to standard-of-care (SOC) therapies **Medical Methodology:** Targeted combination therapy (Allosteric SHP2 Inhibition + PD-1/CDK4/6 Inhibition) **Optimization Target:** Identification of the Maximum Tolerated Dose (MTD) and achievement of Disease Control

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13. Study Procedures & Methodology

Management Pathway: Standard clinical management for advanced solid tumors under Phase I dose-escalation safety protocols. **Education Protocol (HCP):** Site initiation visits focusing on DLT definitions, AE management (especially for combination toxicities), and RECIST v1.1 evaluation. **Education Protocol (Patient):** Informed consent detailing the experimental nature of the combinations, required PK blood draws, and reporting of AEs. **Quality Audit Mechanism:** Continuous safety reviews by the dose-escalation steering committee prior to opening new dose-level cohorts.

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14. Observation & Follow-Up Schedule

Total Duration: Until disease progression, unacceptable toxicity, or study termination. **Onsite Visit Cadence:** Frequent early visits (e.g., Days 1, 8, 15) for PK/PD sampling, transitioning to Day 1 of each subsequent 21/28-day cycle. **Remote Monitoring Frequency:** Between cycles for safety follow-ups as required by institutional standards. **Checklist Review Interval:** Tumor assessments (CT/MRI) performed typically every 8 weeks to determine response.

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| 03-Apr-2026 | | Clinical Operations Lead | [Name] | ___ |** ***03-Apr-2026 | | Lead Statistician | [Name] |*** _____ | 03-Apr-2026 |